

HR ATTRITION PREDICTION DASHBOARD

PROJECT OBJECTIVE

The objective of this project is to analyze employee attrition patterns and identify the key factors that influence employee turnover using Python, Machine Learning, and Power BI.

DATASET DESCRIPTION

The project uses the IBM HR Analytics Employee Attrition dataset containing 1,470 employee records and 35 attributes.

Important attributes include:

- Age
- Attrition
- Department
- Gender
- Monthly Income
- OverTime
- Job Satisfaction
- Work Life Balance
- Years At Company
- Job Role

TOOLS & TECHNOLOGIES USED

- Python
- Pandas
- NumPy
- Matplotlib
- Scikit-Learn
- Power BI
- Jupyter Notebook
- Microsoft Excel

DATA ANALYSIS PERFORMED ATTRITION ANALYSIS

- Calculated overall employee attrition rate.
- Identified departments with higher attrition.
- Analyzed the impact of overtime on employee turnover.
- Studied the relationship between job satisfaction and attrition.
- Evaluated salary and tenure effects on employee retention.

MACHINE LEARNING MODEL

ALGORITHM USED

Random Forest Classifier

Features Used

- Age
- Monthly Income
- Years At Company
- OverTime
- Job Satisfaction
- Work Life Balance

OBJECTIVE

Predict whether an employee is likely to leave the organization.

DASHBOARD PAGES

PAGE 1: HR OVERVIEW

- Total Employees
- Employees Left
- Attrition Rate
- Average Monthly Income
- Attrition Distribution
- Gender Distribution
- Department Distribution

PAGE 2: ATTRITION ANALYSIS

- Overtime vs Attrition
- Job Satisfaction vs Attrition
- Department-wise Attrition

PAGE 3: PREDICTION INSIGHTS

- Feature Importance Analysis
- Business Insights
- Recommendations

KEY FINDINGS

- Attrition Rate = 16.12%
- Employees with low job satisfaction have the highest attrition.
- Overtime significantly increases attrition risk.
- Monthly Income is the strongest predictor of employee turnover.
- Employees with fewer years at the company are more likely to leave.

FEATURE IMPORTANCE RESULTS

Feature	Importance
Monthly Income	34.99%
Age	23.20%
Years At Company	16.82%
OverTime	9.00%
Job Satisfaction	8.38%
Work Life Balance	7.61%

RECOMMENDATIONS

- Improve employee compensation packages.
- Reduce excessive overtime.
- Increase employee engagement programs.
- Focus on retaining new employees.
- Strengthen work-life balance initiatives.

CONCLUSION

This project successfully identified the major factors affecting employee attrition and demonstrated how data analytics and machine learning can support workforce retention strategies.

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